

Mock Practical 1: Specimen Paper 4, Q3

Time: 1 hr 15 min

Name: **SOLUTION** ()

Class: 26S

Date:

This experiment concerns a coupled pendulum system.

Fig. 1.1 shows six pendulums 1, 2, 3, 4, 5 and 6 connected to a common string.

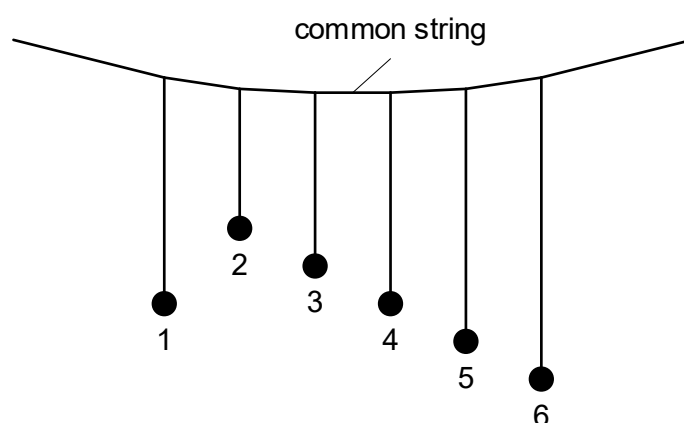


Fig. 1.1

When pendulum 1 is set swinging, the other pendulums move.

You will investigate how the following properties affect the behaviour of a coupled pendulum system.

- the distance between pendulums
- the tension in the common string
- the difference in pendulum lengths
- the mass of the pendulum bobs.

- (a) (i) Set up two pendulums A and B, as shown in Fig. 1.2. Place the loops of the common string on the rods of the clamps. The distance x between the pendulum strings is shown.

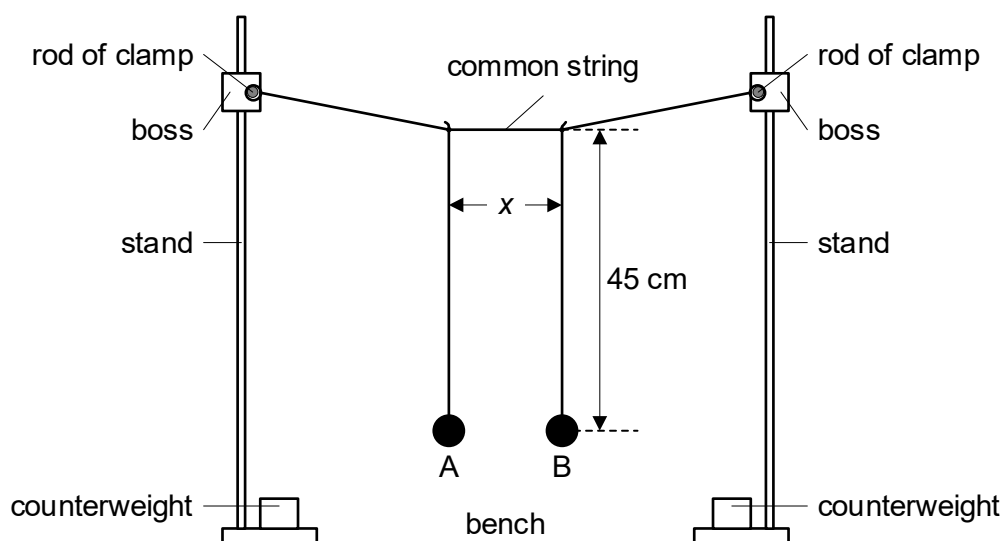


Fig. 1.2

Adjust the distance between the stands so that common string is taut but **not** as taut as possible.

Adjust the knots until each pendulum has a length equal to 45 cm and x is approximately 15 cm.

Measure and record x .

$$x = 15.0 \text{ cm}$$

Pull pendulum A towards you, perpendicular to the plane containing both pendulums. Release the pendulum. Pendulum A will swing then pendulum B will start to swing. After some time A will stop swinging while B carries on swinging.

The time t is the time between releasing A and A stopping for the first time.

Measure and record t .

t_1/s	t_2/s	$\langle t \rangle/\text{s}$
28.05	26.77	27.41

$$t = 27.41 \text{ s}$$

- (ii) Estimate the percentage uncertainty in your value of t .

$$\frac{\Delta t}{\langle t \rangle} = \frac{\frac{1}{2}(28.05 - 26.77)}{\frac{1}{2}(28.05 + 26.77)} \times 100\% = 2.3\% \quad \text{or} \quad \Delta t \approx 1 \text{ s (uncertainty in judging when A stops)}$$

$$\text{percentage uncertainty in } t = 2.3\%$$

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- (iii) It is also possible to determine time t using B. The time $2t$ is the time between releasing A and B stopping for the first time.

Discuss whether this second method of determining t is better or worse than the method used in (a)(i).

$\Delta t \approx 1$ s (uncertainty in judging when A or B stops) is similar for both t and $2t$ measurements.

Since $2t$ is greater, percentage uncertainty for this measurement is smaller and thus better.

- (b) (i) Repeat (a)(i) for at least two more values of x in the range 5 cm to 25 cm.
Tabulate these results. Include the results from (a)(i).

t is time for A to first stop momentarily **OR** $\frac{1}{2}$ (time for B to first stop momentarily).

x/cm	t_1/s	t_2/s	$\langle t \rangle/\text{s}$
8.0	20.05	21.52	20.79
15.0	28.05	26.77	27.41
23.0	34.12	32.58	33.35

Note: use 75% (or 15cm) of the range between 5cm to 25cm.

[2]

- (ii) Comment on the trend in your results.

The time that pendulum A takes to stop for the first time increases as the separation between

the two pendulums increase.

[1]

- (c) In the following experiments you will use the **same value of x** throughout.
Choose **one** value of x from the values in (b)(i) to use in the following experiments.
Record your choice of x .

$x = 8.0$ cm

Explain your choice of x .

Since $t \geq 20$ s for all x , the percentage uncertainty in t is low. $x = 8.0$ cm was chosen as it has the shortest duration, so more repeated timings can be done to reduce percentage uncertainty in t .

OR: $x = 23.0$ cm was chosen because it has the lowest percentage uncertainty in t .

[1]

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- (d) (i) For the arrangement shown in Fig. 1.2, adjust the positions of the pendulums to make x equal to the value chosen in (c).

Gently move the stands apart so that the common string is as taut as possible without the stands falling over. This will increase the tension in the common string.

Measure and record t .

Time for A to first stop momentarily OR $\frac{1}{2}(\text{time for B to first stop momentarily})$, t :

t_1/s	t_2/s	$\langle t \rangle/\text{s}$
64.97	63.86	64.42

$$t = 64.42 \text{ s}$$

- (ii) Comment on the effect on t of increasing the tension in the common string.

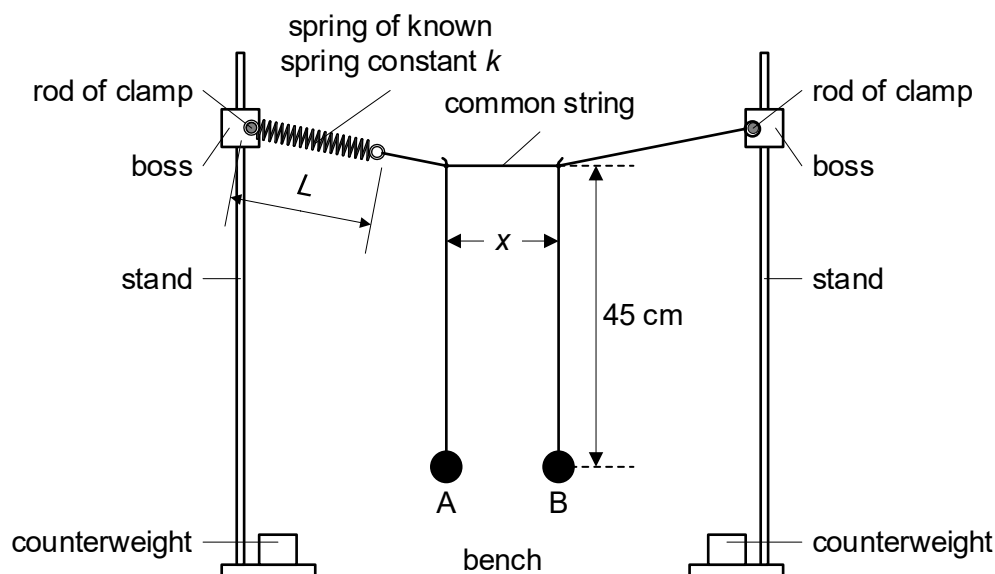
Increasing the tension of the common string results in a longer time for pendulum A to first

come to a stop.

[1]

- (iii) Describe, using a diagram, how you could use a spring to investigate the effect of tension in the common string on t .

You may suggest the use of any additional apparatus commonly found in a school physics laboratory.



1. Measure the natural length L_0 of a spring of known spring constant k using a half metre rule.

2. Set up the apparatus as shown above by adding the spring to one end of the common string and keeping $x = 8.0$ cm and length of pendulum as 45.0 cm throughout the experiment.

3. Measure the length L of the spring using a half meter rule. Extension of the spring is $e = L - L_0$. Use this extension e to determine the tension $T = ke$.

4. Repeat step (a)(i) to determine t using a stopwatch.

5. Repeat steps 2 to 4 for 8 sets of t and T by varying the distance between the retort stands.

6. Plot a graph of t against T to determine the relationship between t and T . [4]

Note

[1]: diagram showing spring connected to common string at one end, not between pendulums.

[1]: measure time t with stopwatch. Measure lengths L , L_0 with ruler. Determine tension T .

[1]: repeat for 8 sets by varying distance between retort stands to change tension.

[1] plot t vs T to study effects of tension on time; accept proposed power relationship $t = kT^n$.

- (e) Keep x equal to the value chosen in (c).

The distance y is the difference in length between the two pendulums, as shown in Fig. 1.3.

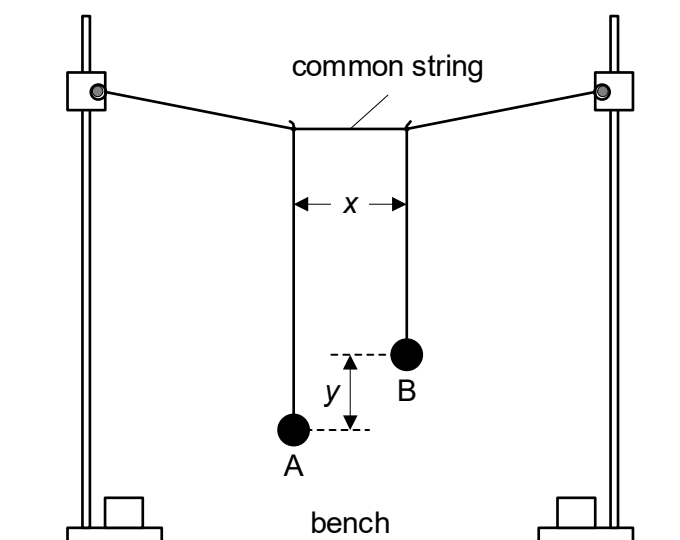


Fig. 1.3

- (i) Reduce the length of pendulum B until y is approximately 4 cm.
Measure and record y and t .

time for B to first stop momentarily, $2t$:

$(2t_1)/s$	$(2t_2)/s$	$\langle 2t \rangle /s$	t/s
20.33	21.99	21.16	10.58

$$y = 4.4 \text{ cm}$$

$$t = 10.58 \text{ s} \quad [1]$$

- (ii) Reduce the length of pendulum B until y is approximately 8 cm.
Measure and record y and t .

time for B to first stop momentarily, $2t$:

$(2t_1)/s$	$(2t_2)/s$	$\langle 2t \rangle /s$	t/s
14.14	13.47	13.81	6.903

$$y = 8.0 \text{ cm}$$

$$t = 6.903 \text{ s (4 s.f.)} \quad [1]$$

- (iii) It is suggested that

$$t = \frac{k}{y}$$

where k is a constant.

Use your values from (e)(i) and (e)(ii) to determine two values of k .

Values of k

$$k_1 = y_1 t_1 = 4.4 \times 10.58 = 46.55 = 47 \text{ cm s (2 s.f.)}$$

$$k_2 = y_2 t_2 = 8.0 \times 6.903 = 55.22 = 55 \text{ cm s (2 s.f.)}$$

$$\text{first value of } k = 47 \text{ cm s}$$

$$\text{second value of } k = 55 \text{ cm s} \quad [2]$$

- (iv) Justify the number of significant figures given in your values of k .

Since values of y are 2 s.f., and values of t are 4 s.f., k follows the least s.f. which is 2 s.f. [1]

The effect of the difference in length y between the pendulums and the validity of the relationship $t = k/y$ were investigated. This was well answered with many successfully justifying, or otherwise, the relationship and using an appropriate number of significant figures for k .

- (v) State whether the results of your experiment support the relationship suggested in (e)(iii). Justify your conclusion by referring to your answers in (a)(ii).

$$\text{Percentage uncertainty in } k = \frac{\Delta k}{\langle k \rangle} = \frac{\frac{1}{2}(k_2 - k_1)}{\frac{1}{2}(k_2 + k_1)} = \frac{\frac{1}{2}(55 - 47)}{\frac{1}{2}(55 + 47)} \times 100\% = 7.8\%$$

Since the percentage uncertainty for k is 7.8% is greater than the percentage uncertainty in a(ii) of 2.3%, the results do not support the suggested relationship in (e)(iii). [1]

- (f) Fig. 1.4 shows two pendulums of equal length. The pendulum bobs consist of a number of slotted masses.

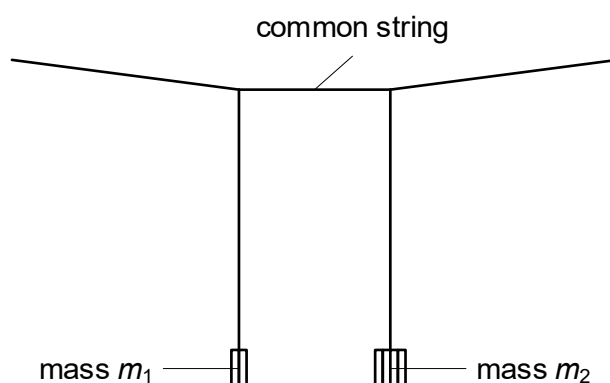


Fig. 1.4

In an investigation, the mass m_1 is fixed at 40 g. The mass m_2 is varied. Table 1.1 shows data for m_2 and t .

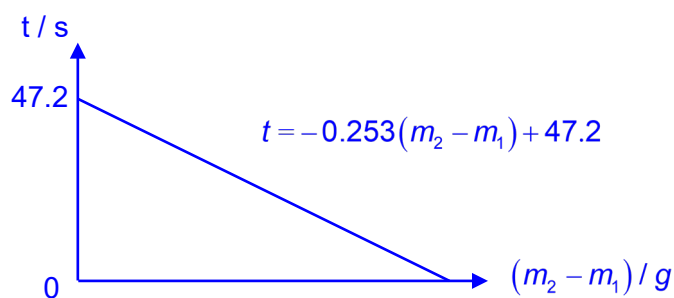
m_2/g	80	100	120	140	160
t/s	37.78	31.07	26.86	22.33	16.88
$(m_2 - m_1)/\text{g}$	40	60	80	100	120

Table 1.1

Complete Table 1.1.

Plot a suitable graph using a spreadsheet to determine t when the two masses are equal.

Sketch the graph in the space below. Write down the equation of the trendline.



Graph: $t = k(m_2 - m_1) + c$

where k is the gradient and c is the vertical intercept.

When the 2 masses are equal, $t = c$.

$$t = \underline{47.2 \text{ s}} \quad [4]$$

[24 marks]

End of Paper